# LLM–Guided Label Propagation with Hypergraph for Drug Repurposing

Yuqing Qian[1], Yijie Ding[2] and Quan Zou[1,*]

[1] The Institute of Fundamental and Frontier Sciences, University of Electronic Science and Technology of China, Chengdu, 611731, P. R. China
[2] Faculty of Computing, Harbin Institute of Technology, Harbin, 150001, P. R. China
zouquan@nclab.net

**Abstract.** Drug repositioning aims to discover new therapeutic indications for existing drugs. Although numerous computational methods have been developed to predict drug and disease associations (DDAs), existing approaches still face several major challenges. These include limited predictive ability for previously unseen diseases, incomplete integration of multi-source biomedical data, and the inability of traditional graph structures to effectively capture higher-order relationships among biomedical entities. To overcome these limitations, we introduce LLM-GLPH, a framework for drug repositioning. First, LLM generates biomedical descriptions of drugs and diseases based solely on their entity names and prompts. The generated biotexts are then encoded into dense embedding vectors to obtain unified semantic representations. Next, the Weighted K Nearest Known Neighbors (WKNKN) algorithm is used to alleviate data sparsity and improve the prediction of new diseases. Furthermore, a hypergraph structure is constructed to model higher-order relationships between drugs and diseases, and the hypergraph Laplacian is incorporated into a graph-based label propagation framework to infer potential DDAs. Experiments performed on three public datasets show that our method achieves competitive and robust results under both standard association prediction and de novo disease prediction settings. Further validation through case studies on breast neoplasms supports the practical effectiveness of the proposed approach.

**Keywords:** Drug repositioning, Large language models, Hypergraph learning, Label propagation

## 1 Introduction

Drug repositioning refers to the identification of new therapeutic uses for existing drugs beyond their original clinical indications [1]. Compared with traditional de novo drug discovery, drug repositioning offers significant advantages, including reduced development time, lower cost, and improved safety profiles, since repurposed drugs have often already passed substantial preclinical and clinical evaluations. In recent years, drug repositioning has emerged as an efficient strategy to address urgent medical needs.

These needs include particularly complex diseases, rare disorders, and emerging infectious diseases.

Biological systems are highly complex. As a result, systematically identifying promising drug candidates via experimental approaches is often impractical. These challenges have motivated the development of computational methods to accelerate and prioritize drug repositioning efforts [2].

One important category of approaches is collaborative filtering (CF)-based methods [3]. These methods aim to recover missing associations by decomposing the DDA matrix into a low-dimensional latent representation. For example, CDPMF [4] combines probabilistic matrix factorization with multi-view contrastive learning to capture rich relationships among drugs and diseases. Similarly, AutoDR [3] employs collaborative metric learning to learn latent embeddings of drugs and diseases in a shared metric space to estimate their association probabilities.

Driven by recent progress in deep learning, graph neural networks (GNN) based methods have been extensively explored for drug repositioning due to their ability to represent intricate interactions among biomedical entities in heterogeneous networks [5]. DRHGCN [6] leverages graph convolutional operations to extract both inter- and intra-domain features from drug similarity, disease similarity, and DDA networks. HDGAT [5] further improves representation learning by introducing hierarchical and dynamic attention mechanisms to capture multi-level structural information. Other approaches integrate neural collaborative filtering and graph learning techniques, such as DRWBNCF and DRGBCN [7].

Despite the promising progress achieved by the existing computational methods, three significant challenges remain in drug repositioning research.

First, effectively integrating drug and disease representations often requires multi-view information derived from multiple data sources [8]. However, heterogeneous databases differ in identifiers, naming conventions, and annotation standards, making entity alignment difficult and often causing incomplete feature representations [2]. Such incompleteness can weaken learning representations and limit drug repositioning models' performance.

Second, most existing approaches overlook drug association prediction for new diseases, which leads to a cold-start problem in drug repositioning [8]. CF-based methods depend on historical DDA matrices, while GNN-based methods rely on existing graph edges for message passing. When a new disease has no or few known associations, both methods struggle to learn reliable representations, leading to poor prediction performance.

Third, many existing methods construct drug and disease similarity networks as graphs to improve DDA prediction [9]. However, traditional graphs mainly capture pairwise relationships between two nodes, while biomedical systems often involve group-level interactions or shared functional mechanisms among multiple drugs or diseases. This limitation makes it difficult to model higher-order relationships among multiple entities.

To address these challenges, we proposed a novel method, called **LLM**–**G**uided **L**abel **P**ropagation with **H**ypergraph (LLM-GLPH). The main contributions to our work are summarized as follows:

1. We leveraged large language models (LLMs) to generate unified semantic representations of drugs and diseases from entity names. The generated biomedical descriptions were encoded into dense embeddings to obtain informative drug and disease features.
2. We incorporated the WKNKN algorithm to estimate potential associations and reduce data sparsity. This helps infer candidate DDAs for new diseases with limited known associations.
3. We introduced a hypergraph structure to capture high-order interactions that may exist across groups of drugs and diseases. The Hypergraph Laplacian was then used with graph-based label propagation to infer potential DDAs.
4. We conducted comprehensive experiments on three benchmark datasets, and the results verify that our method outperforms representative baselines in both association prediction and de novo disease prediction settings. Case studies on Breast Neoplasms further confirm that the predicted drugs are supported by biomedical evidence.

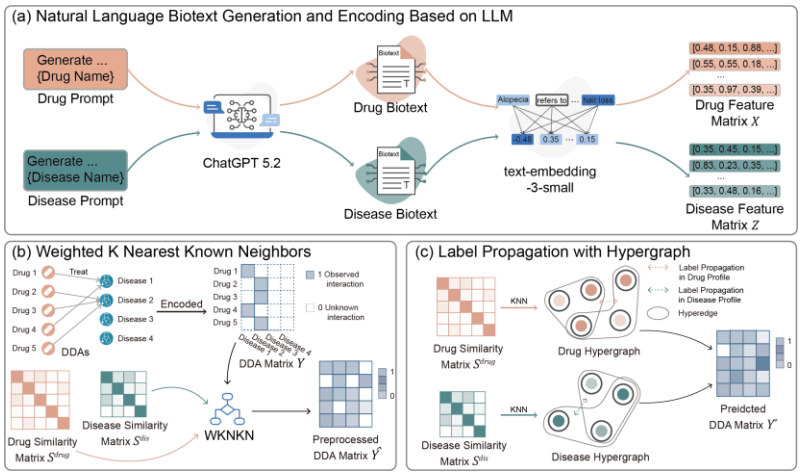



**Fig. 1.** Overview of the proposed LLM-GLPH framework for drug repurposing.

## Materials and Methods

This section describes the datasets used in this study and introduces the proposed framework for DDA prediction. The overall procedure of our method is shown in Fig. 1.

### 1.1 Datasets

To evaluate LLM-GLPH's performance on drug repurposing, we conducted experiments on real-world benchmark datasets. Specifically, we adopted widely recognized gold-standard datasets, F [10], C [11], and LRSSL [12].

### 1.2 Natural Language Biotext Generation and Encoding Based on LLM

This study employed one of the most popular LLMs, ChatGPT 5.2, to generate the biotext of the drugs and diseases. ChatGPT 5.2 is developed based on the transformer architecture and learns from large-scale, diverse corpora encompassing scientific literature, biomedical texts, technical documents, and general-domain knowledge [13]. For each drug or disease, we constructed a prompt template containing the entity name and an instruction requesting a concise biomedical explanation. The prompts were submitted to the LLM through the API[1] interface to generate descriptive biotext for each entity. By leveraging its pre-trained biomedical and general-domain knowledge, ChatGPT synthesizes relevant information such as mechanisms of action, therapeutic uses, molecular targets (for drugs), and pathological characteristics or related biological processes (for diseases).

We employed text-embedding-3-small, developed by OpenAI [13]. This pretrained model projects textual inputs into dense semantic representations and outputs 1536-dimensional feature vectors. With improvements in efficiency and semantic modeling capability, it can provide effective textual representations for biomedical entities and related downstream tasks. The feature representations of drugs and diseases are generated by the large language model.

Let $X \in R^{n\times 1536}$ and $Z \in R^{m\times 1536}$ denote the drug feature matrix and denote the disease feature matrix, where each row corresponds to a 1536-dimensional embedding vector obtained from the generated biotext, $n$ and $m$ represent the numbers of drugs and diseases, respectively.

### 1.3 Weighted K Nearest Known Neighbors

The DDA matrix is extremely sparse because only a small fraction of all possible associations has been experimentally validated. To alleviate these issues, the WKNKN algorithm is employed as a preprocessing strategy to estimate the likelihood of potential associations [14].

Let's $Y \in \mathbb{R}^{n\times m}$ denote the DDA matrix. Specifically, $Y_{i,j} = 1$ indicates that the drug $i$ is associated with disease $j$, and $Y_{i,j} = 0$ otherwise. Based on these embedding representations, we constructed similarity matrices for drugs and diseases to capture their semantic relationships. The disease similarity matrix $S^{dis} \in \mathbb{R}^{m\times m}$ is computed using:

$$S_{i,j}^{dis} = \exp\left(-\frac{\|Z_{i,:}-Z_{j,:}\|_F^2}{\sigma^2}\right). \quad (1)$$

where $\sigma$ is the kernel bandwidth parameter, and the default value is 0.5.

For a given disease $i$, the algorithm first identifies its $k$ nearest known neighbors according to the disease similarity matrix $S^{dis}$:

[1] https://platform.openai.com/

$$knn = KNKN\ (i, k, S^{dis}), \tag{2}$$

where $KNKN()$ returns the $k$ nearest diseases that ranked in descending order of similarity. The association profile of disease $i$ is then estimated as a weighted combination of the association profiles of its neighbors:

$$\hat{Y}_{i,:} = \frac{1}{Z_i}\sum_{j=1}^{k} w_j Y_{knn_j,:}, \tag{3}$$

where $Z$ is a normalization term defined as:

$$Z_i = \sum_{j=1}^{k} S_{i,knn_j}^{dis}. \tag{4}$$

The weight assigned to each neighbor decreases exponentially with its rank:

$$w_j = \mu^{j-1} Y_{knn_j,:}, \tag{5}$$

where $\mu$ is a decay term controlling the contribution of lower-ranked neighbors, and it is set to $\mu = 0.9$.

Finally, the updated association matrix is obtained by preserving the known associations while incorporating the estimated scores:

$$\hat{Y} = \max(\hat{Y}, Y). \tag{6}$$

It is worth noting that WKNKN selects the $k$ nearest known neighbors rather than simply the $k$ nearest neighbors. This design ensures that the neighbors used for inference contain informative association profiles.

### 1.4 Label Propagation with Hypergraph

In a traditional graph, an edge connects only two vertices, which limits the ability to model higher-order relationships among multiple entities [15]. However, in biomedical systems, drugs or diseases often exhibit group-wise similarities, where one entity may simultaneously relate to multiple similar entities [16]. Modeling such relationships using only pairwise edges may lead to information loss. In contrast, a hypergraph allows each hyperedge to connect multiple vertices simultaneously, enabling higher-order structural relationships. Therefore, compared with ordinary graphs, hypergraphs can better capture the complex similarity of structures among drugs or diseases.

Let $G = (V, E)$ denote a hypergraph, $V$ is a finite set of vertices $v$, $E$ is a set of hyperedges, where each hyperedge $e \in E$ connects multiple vertices. We used matrix $H \in \mathbb{R}^{|V|\times|E|}$ to represent hypergraph $G$ where entries $H_{v,e} = 1$ if $v \in e$ and 0 otherwise. To construct the hypergraph structure for diseases, we adopt a k-nearest neighbor (KNN) strategy based on the disease similarity matrix. Specifically, for a given disease $i$, we first identify its $k$ nearest neighbors according to Eq. (2). Then, a hyperedge is formed by connecting disease $i$ together with its $k$ nearest neighbors. The weights of all hyperedges are set to 1. By repeating this process for all diseases, a set of hyperedges is obtained, forming the disease hypergraph.

Let's $D \in \mathbb{R}^{|V|\times|V|}$ and $F \in \mathbb{R}^{|E|\times|E|}$ are the diagonal matrices, it calculated by

$$D_{i,i} = \sum_{e \in E} H_{i,e}, \tag{7}$$

and

$$F_{j,j} = \sum_{v \in V} H_{v,j}. \tag{8}$$

Then, the hypergraph Laplacian is calculated as follows:

$$L^{dis} = I_m - D^{-0.5} H F^{-1} H^T D^{-0.5}, \tag{9}$$

where $I_m$ is the $m \times m$ identity matrix.

Similarly, the hypergraph structure for drugs can be constructed using the drug similarity matrix and the same KNN strategy. The corresponding drug hypergraph Laplacian is denoted as $L^{drug}$.

Following the guilt-by-association principle, we employ a graph-based label propagation (GLP) method. The GLP algorithm propagates label information through graph structures, allowing the prediction of labels for both labeled and unlabeled data. Let's $Y^*$ denote the predicted association matrix. The objective function of GLP is formulated as:

$$\text{argmin}_{Y^*} \frac{1}{2} \left\| Y^* - \hat{Y} \right\|_F^2 + \frac{\lambda_1}{2} \text{Tr}\left( Y^{*T} L^{drug} Y^* \right) + \frac{\lambda_2}{2} \text{Tr}\left( Y^* L^{dis} Y^{*T} \right), \tag{10}$$

where $\|\cdot\|_F$ denotes the Frobenius norm and $Tr(\cdot)$ denotes the trace of a matrix. The first term is a reconstruction loss that encourages the predicted $Y^*$ to remain consistent with the preprocessed association matrix $\hat{Y}$. The second term imposes a smoothness constraint on the drug graph, encouraging drugs with high similarity to have similar association patterns with diseases. Similarly, the third term enforces smoothness over the disease graph, ensuring that diseases with similar characteristics tend to share similar drug associations. The hyperparameters $\lambda_1 > 0$ and $\lambda_2 > 0$ are regularization coefficients that balance the contributions of these terms.

By setting the derivative of Eq. (10) with respect to $Y^*$ to zero, we obtain:

$$(I + \lambda_1 L^{drug}) Y^* + \lambda_2 \hat{Y} L^{dis} = Y^*. \tag{11}$$

Eq. (11) is a Sylvester equation, which can be efficiently solved using the Schur decomposition method.

## Experimental and Results

### 1.5 Baselines

To assess the effectiveness of LLM-GLPH, a comparative analysis is conducted between LLM-GLPH and eight advanced approaches across three gold-standard datasets. Included in these baseline approaches are four models (SCPMF [17], DRWBNCF [18], CDPMF [4], and AutoDR [3]) that rely on CF and four models (DRHGCN [6], HCCF [19], HDGAT [5], and DRGBCN [7]) utilizing GNNs.

### 1.6 Evaluation Metrics

In this study, we employed two widely used evaluation metrics, namely the AUPR and AUC. Both metrics are commonly used in DDA prediction because they reflect the discriminatory capabilities of prediction models.

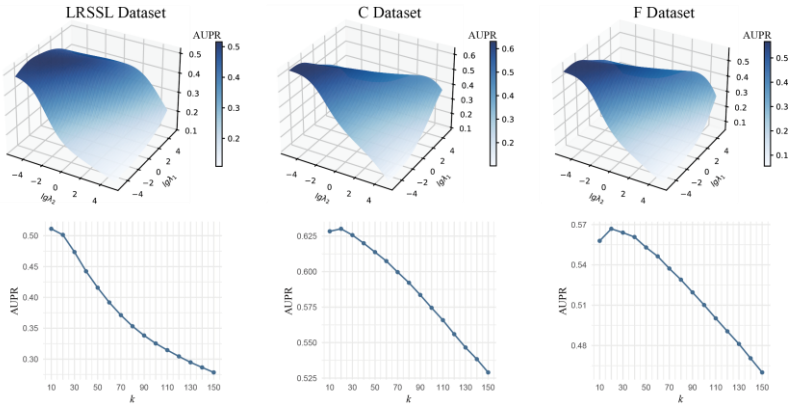



**Fig. 2.** Hyperparameter sensitivity analysis on the LRSSL, C, and F datasets.

### 1.7 Hyperparameter Setting

In our method, three hyperparameters, namely $\lambda_1$, $\lambda_2$, and $k$, were tuned using a grid search strategy. The optimal hyperparameter settings were determined based on the best AUPR performance. Specifically, the search space for $\lambda_1$ and $\lambda_2$ ranges from $2^{-5}$ to $2^5$ with a step size of $2^1$. The hyperparameter $k$ was searched in the range from 10 to 50 with an interval of 5. The sensitivity of the model with respect to these hyperparameters on different datasets is illustrated in Fig. 2.

### 1.8 Ablation Analysis

Ablation experiments were carried out to examine the role of each module, where different components were removed separately from the complete framework. Specifically, three variants were considered: (1) w/o HG, which removes the hypergraph structure and replaces it with a conventional graph; (2) w/o LLM, which excludes the LLM-generated semantic representations and instead uses the multi-view features constructed following Ai et al.'s work [20]; and (3) w/o WKNKN, which removes the WKNKN preprocessing step for handling sparsity in the DDA matrix.

Fig. 3 presents the AUPR curves of the proposed method and its ablated variants on different datasets under both association prediction and de novo disease prediction settings. As shown in the figure, the full model consistently achieves the best performance compared with all ablated versions. Excluding the LLM-based biotext representations leads to a noticeable performance degradation. In the w/o LLM variant, LLM semantic representations are replaced with multi-view features constructed following Ai et al.'s

work [20]. However, such multi-view features are often incomplete because they rely on manually curated biological data from multiple sources.

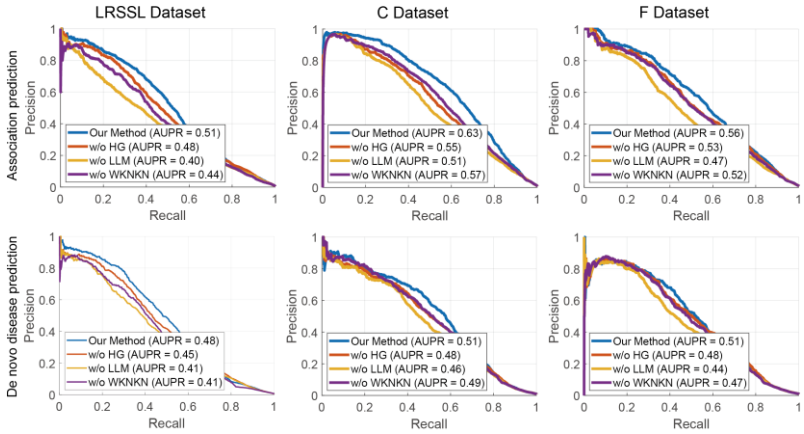



**Fig. 3.** Precision–Recall curves of the ablation study under two prediction settings. The first row shows the results for association prediction, while the second row presents the performance for de novo disease prediction.

## 1.9 Performance Under Association Prediction Setting

**Table 1.** Comparison of LLM-GLPH (Our) and Other Methods in 10-CV under association prediction setting.

| Methods | LRSSL | | C | | F | |
|---|---|---|---|---|---|---|
| | AUPR | AUC | AUPR | AUC | AUPR | AUC |
| SCPMF | 0.27±0.002 | 0.89±0.001 | 0.42±0.004 | 0.91±0.002 | 0.34±0.006 | 0.89±0.001 |
| DRWBNCF | 0.34±0.005 | 0.93±0.001 | 0.56±0.004 | 0.94±0.001 | 0.49±0.001 | 0.92±0.002 |
| CDPMF | 0.35±0.005 | 0.94±0.005 | 0.60±0.005 | 0.95±0.005 | 0.55±0.005 | 0.95±0.003 |
| AutoDR | 0.40±0.043 | 0.95±0.010 | 0.63±0.043 | 0.96±0.009 | **0.57±0.041** | 0.95±0.005 |
| DRHGCN | 0.38±0.022 | 0.96±0.006 | 0.58±0.035 | 0.96±0.005 | 0.49±0.041 | 0.95±0.011 |
| HCCF | 0.36±0.031 | 0.89±0.011 | 0.56±0.031 | 0.91±0.012 | 0.48±0.036 | 0.90±0.010 |
| HDGAT | 0.40±0.005 | 0.94±0.001 | 0.57±0.008 | 0.94±0.001 | 0.51±0.005 | 0.92±0.002 |
| DRGBCN | 0.26±0.006 | 0.94±0.001 | 0.45±0.006 | 0.95±0.001 | 0.41±0.007 | 0.93±0.003 |
| Our | **0.50±0.003** | **0.97±0.000** | **0.63±0.003** | **0.97±0.001** | 0.56±0.003 | **0.96±0.001** |

To comprehensively evaluate the effectiveness of the proposed model in predicting potential DDAs, we compared our method with several state-of-the-art approaches in the association prediction setting. Specifically, a 10-fold cross-validation (10-CV) strategy was adopted, where known DDAs were randomly divided into ten subsets. In each fold, nine subsets were used for training and the remaining one for testing. All unknown

associations were treated as candidate pairs to be predicted. The results are summarized in Table 1.

Overall, the proposed approach demonstrates the most competitive performance across different datasets and metrics. On LRSSL, it achieves an AUPR value of 0.50 and an AUC value of 0.97, which are higher than those of the strongest baseline methods, including HDGAT with an AUPR of 0.40 and DRHGCN with an AUC of 0.96. Compared with classical CF-based methods such as SCPMF and CDPMF, our approach improves the AUPR by a large margin. In addition, compared with recent GNN-based approaches such as AutoDR and DRHGCN, LLM-GLPH still demonstrates clear advantages, highlighting its superior capability in modeling complex DDAs.

### Performance Under De Novo Disease Prediction Setting

We further assessed the ability of the proposed LLM-GLPH framework to identify candidate drugs for previously unseen diseases. Specifically, a 10-CV was adopted under the de novo disease prediction setting, where diseases were randomly partitioned into ten subsets. In each fold, the associations involving diseases from one subset were withheld as the test set, while the remaining diseases and their known associations were used for model training. The comparative results are reported in Table 2.

**Table 2.** Comparison of LLM-GLPH (Our) and Other Methods in Predicting Potential Drugs for New Diseases.

| Methods | LRSSL Dataset | | C Dataset | | F Dataset | |
|---|---|---|---|---|---|---|
| | AUPR | AUC | AUPR | AUC | AUPR | AUC |
| SCPMF | 0.03±0.001 | 0.77±0.003 | 0.07±0.001 | 0.74±0.001 | 0.06±0.001 | 0.71±0.002 |
| DRWBNCF | 0.05±0.001 | 0.76±0.004 | 0.23±0.008 | 0.82±0.005 | 0.21±0.003 | 0.81±0.003 |
| CDPMF | 0.30±0.002 | 0.84±0.003 | 0.33±0.006 | 0.87±0.001 | 0.31±0.006 | 0.87±0.002 |
| AutoDR | 0.23±0.002 | 0.89±0.006 | 0.40±0.004 | 0.89±0.004 | 0.29±0.001 | 0.86±0.002 |
| DRHGCN | 0.06±0.001 | 0.85±0.002 | 0.20±0.002 | 0.83±0.002 | 0.11±0.001 | 0.79±0.003 |
| HCCF | 0.01±0.000 | 0.55±0.002 | 0.23±0.007 | 0.82±0.003 | 0.21±0.005 | 0.81±0.004 |
| HDGAT | 0.14±0.001 | 0.75±0.004 | 0.16±0.001 | 0.74±0.001 | 0.12±0.001 | 0.73±0.002 |
| DRGBCN | 0.01±0.000 | 0.56±0.003 | 0.01±0.000 | 0.49±0.001 | 0.01±0.001 | 0.53±0.006 |
| Our | **0.50±0.005** | **0.94±0.001** | **0.51±0.007** | **0.95±0.001** | **0.50±0.004** | **0.94±0.001** |

As indicated by the experimental results, LLM-GLPH maintains the strongest overall predictive performance on all benchmark datasets and evaluation metrics. In particular, on the LRSSL dataset, it obtains an AUPR value of 0.50 and an AUC value of 0.94, outperforming all competing methods by a considerable margin. The second-best AUPR is achieved by CDPMF (AUPR=0.30), while the best AUC among the baselines is achieved by AutoDR (AUC=0.89). This demonstrates that our model can more effectively identify potential drug–disease associations, especially under highly sparse

conditions. Similar trends are observed in the C dataset and the F dataset. In these datasets, LLM-GLPH achieves AUPR values of 0.51 and 0.50, and AUC values of 0.95 and 0.94, respectively, outperforming all competing approaches by a clear margin.

**Table 3.** Top-Ranked Candidate Drugs Predicted for Breast Neoplasms based on the F Dataset.

| Ranks | DrugBank IDs | Candidate Drugs | PubMed ID |
|---|---|---|---|
| 1 | DB01196 | Estramustine | - |
| 2 | DB00882 | Clomifene | 27580430, 25751625 |
| 3 | DB00655 | Estrone | 35191604, 17261762 |
| 4 | DB00515 | Cisplatin | 31194954 |
| 5 | DB01406 | Danazol | 6697011 |

## 1.10 Case Study

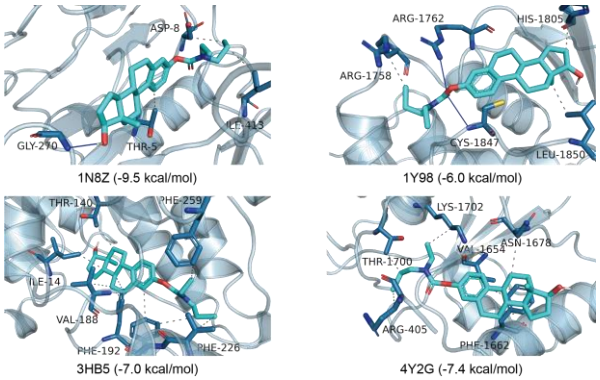



**Fig. 4.** Molecular docking of Estramustine with the target protein.

To evaluate the effectiveness of LLM-GLPH in identifying potential therapeutic drugs, we analyzed the top five predicted candidate drugs for Breast neoplasms based on the F dataset. Table 3 shows that most candidate drugs predicted by our method can be supported by existing biomedical literature. The model predicted Estramustine as the top-ranked candidate drug. Interestingly, no direct literature evidence was found linking Estramustine to Breast neoplasms, suggesting that it may represent a novel drug repositioning candidate.

To further investigate the potential interaction between Estramustine and disease-related target proteins, molecular docking analysis was performed using AutoDock Vina [21]. As shown in Fig. 4, Estramustine binds stably in the active pockets of multiple target proteins. This suggests that Estramustine may effectively interact with proteins associated with breast neoplasm-related biological processes.

## 2 Conclusion and Discussion

Existing computational methods for drug repurposing often rely on multi-view biological features collected from multiple heterogeneous data sources, whose integration is inherently complex and incomplete. In addition, many methods struggle with the cold-start problem when predicting drug associations for previously unseen diseases. Furthermore, most existing studies demonstrate drug and disease similarities using traditional graphs. Traditional graph-based methods are generally designed to describe pairwise dependencies, making it difficult to fully characterize high-order interactions among biomedical entities.

To address these challenges, we introduced the LLM-GLPH framework. Specifically, LLMs are used to generate biomedical biotexts for drugs and diseases and encode them into unified semantic embeddings. The WKNKN algorithm alleviates the sparsity of the DDA matrix and improves prediction capability for new diseases. Moreover, hypergraph structures are constructed to capture higher-order relationships among biomedical entities, and the resulting hypergraph Laplacian is incorporated into GLP.

Experiments on three benchmark datasets show that the proposed method consistently outperforms several advanced baseline approaches in both association prediction and de novo disease prediction tasks. Case studies on Breast neoplasms further confirm the practical utility of the proposed framework.

**Acknowledgments.** This work is supported in part by the National Natural Science Foundation of China (NSFC 62425107, 62501426), the Zhejiang Provincial Natural Science Foundation of China (Grant No. LY23F020003), the Municipal Government of Quzhou (Grant No.2024D001).

**Disclosure of Interests.** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## References

1. Chen, Y., Z. Wang, J. Wang, Y. Chu, Q. Zhang, Z. Alan, Li,X. Zeng: Self-supervised learning in drug discovery. Science China Information Sciences. **68** (2025)
2. De la Fuente, J., G. Serrano, U. Veleiro, M. Casals, L. Vera, M. Pizurica, N. Gómez-Cebrián, L. Puchades-Carrasco, A. Pineda-Lucena,I. Ochoa: Towards a more inductive world for drug repurposing approaches. Nature Machine Intelligence. **7**(3), 495-508 (2025)
3. Wang, Y., Y. Meng, C. Zhou, X. Tang, P. Zeng, C. Pan, Q. Zhu, B. Zhang,J. Xu: Automatic collaborative learning for drug repositioning. Engineering Applications of Artificial Intelligence (Jan. Pt.B), **139** (2025)
4. Tang, X., Y. Hou, Y. Meng, Z. Wang, C. Lu, J. Lv, X. Hu, J. Xu,J. Yang: CDPMF-DDA: contrastive deep probabilistic matrix factorization for drug-disease association prediction. BMC Bioinformatics. **26**(1), (2025)
5. Huang, S., M. Wang, X. Zheng, J. Chen,C. Tang: Hierarchical and Dynamic Graph Attention Network for Drug-Disease Association Prediction. Journal on Biomedical and Health Informatics (J-BHI). **28**(4), 12 (2024)

6. Lijun, C., L. Changcheng, X. Junlin, M. Yajie, W. Peng, F. Xiangzheng, Z. Xiangxiang,S. Yansen: Drug repositioning based on the heterogeneous information fusion graph convolutional network. Briefings in Bioinformatics (2021)
7. Tang, X., C. Zhou, C. Lu, Y. Meng, J. Xu, X. Hu, G. Tian,J. Yang: Enhancing Drug Repositioning Through Local Interactive Learning With Bilinear Attention Networks. IEEE Journal of Biomedical and Health Informatics. **29** 1644-1655 (2023)
8. Qian, Y., Y. Wang, J. Liu, Q. Zou, Y. Ding, X. Guo,W. Ding: A survey on multi-view fusion for predicting links in biomedical bipartite networks: Methods and applications. Information Fusion. **117** 102894 (2025)
9. Qian, Y., X. Zhang, Y. Wang, Q. Zou, C. Cao, Y. Ding,X. Guo: DTI-RME: a robust and multi-kernel ensemble approach for drug-target interaction prediction. BMC Biology. **23** (2025)
10. Gottlieb, A., G.Y. Stein, E. Ruppin,R. Sharan: PREDICT: a method for inferring novel drug indications with application to personalized medicine. Molecular systems biology. **7**(1), MSB201126 (2011)
11. Luo, H., J. Wang, M. Li, J. Luo, X. Peng, F.-X. Wu,Y. Pan: Drug repositioning based on comprehensive similarity measures and bi-random walk algorithm. Bioinformatics. **32**(17), 2664-2671 (2016)
12. Liang, X., P. Zhang, L. Yan, Y. Fu, F. Peng, L. Qu, M. Shao, Y. Chen,Z. Chen: LRSSL: predict and interpret drug–disease associations based on data integration using sparse subspace learning. Bioinformatics. **33**(8), 1187-1196 (2017)
13. Achiam, J., S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F.L. Aleman, D. Almeida, J. Altenschmidt, S. Altman,S. Anadkat: Gpt-4 technical report. arXiv preprint arXiv:2303.08774 (2023)
14. Ezzat, A., P. Zhao, M. Wu, X. Li,C.K. Kwoh: Drug-Target Interaction Prediction with Graph Regularized Matrix Factorization. IEEE/ACM Transactions on Computational Biology & Bioinformatics. PP(99), 646-656 (2016)
15. Chai, L., L. Tu, X. Wang,Q. Su: Hypergraph modeling and hypergraph multi-view attention neural network for link prediction. Pattern Recognition. **149** 110292 (2024)
16. Jin, Z., X. Zheng, H. Zhou, C. Ji, S. Xiang,C. Tang: DD-HGNN$^+$: Drug-Disease Association Prediction via General Hypergraph Neural Network With Hierarchical Contrastive Learning and Cross Attention Learning. IEEE Journal of Biomedical and Health Informatics. **29** 7810-7819 (2025)
17. Meng, Y., M. Jin, X. Tang,J. Xu: Drug repositioning based on similarity constrained probabilistic matrix factorization: COVID-19 as a case study. Applied Soft Computing (2021)
18. Meng, Y., C. Lu, M. Jin, J. Xu, X. Zeng,J. Yang: A weighted bilinear neural collaborative filtering approach for drug repositioning. Briefings in Bioinformatics (2022)
19. Xia, L., C. Huang, Y. Xu, J. Zhao, D. Yin,J.X. Huang: Hypergraph Contrastive Collaborative Filtering. In: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieva. ACM, Madrid (2022)
20. Ai, C., H. Yang, Y. Ding, J. Tang,F. Guo: Low Rank Matrix Factorization Algorithm Based on Multi-Graph Regularization for Detecting Drug-Disease Association. IEEE/ACM Transactions on Computational Biology and Bioinformatics. **20** 3033-3043 (2023)
21. Bugnon, M., U.F. Röhrig, M. Goullieux, M.A. Perez, A. Daina, O. Michielin,V. Zoete: SwissDock 2024: major enhancements for small-molecule docking with Attracting Cavities and AutoDock Vina. Nucleic acids research. **52**(W1), W324-W332 (2024)